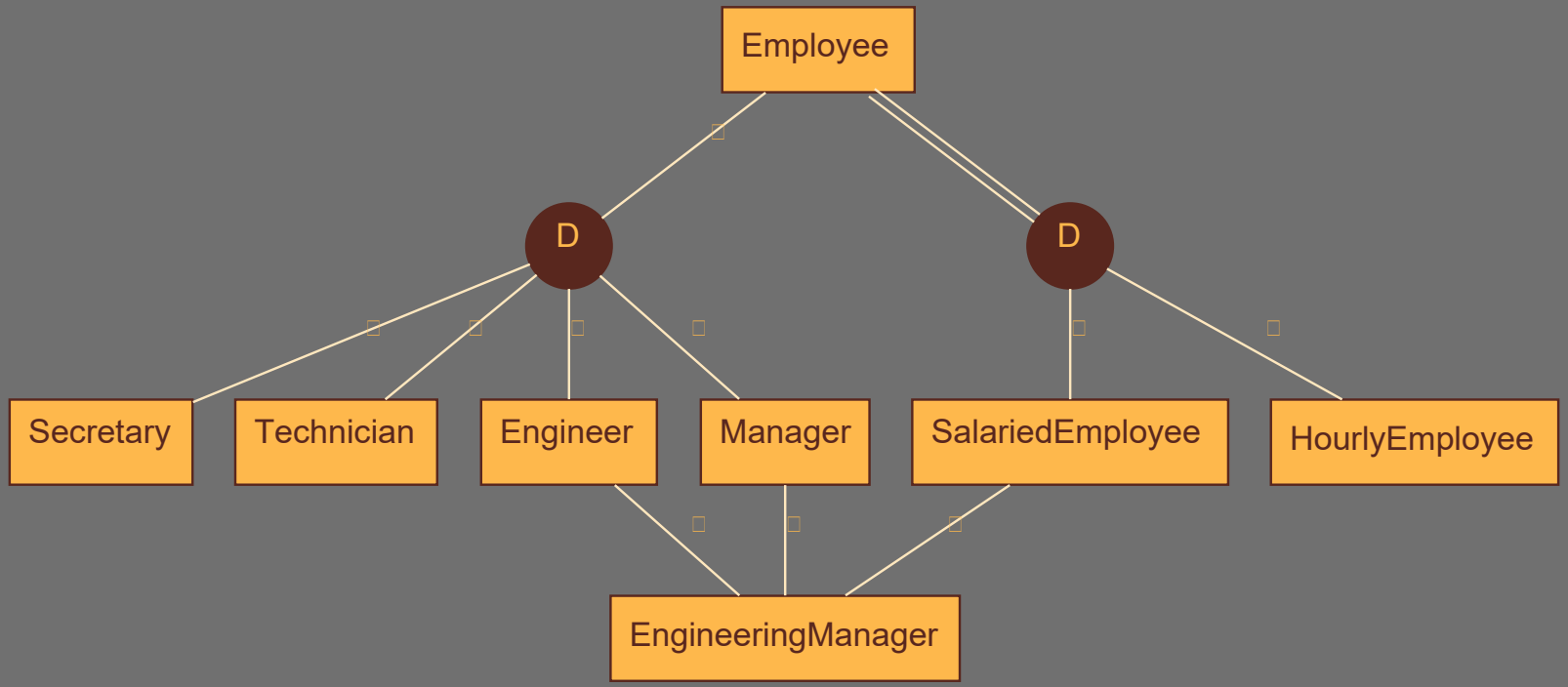
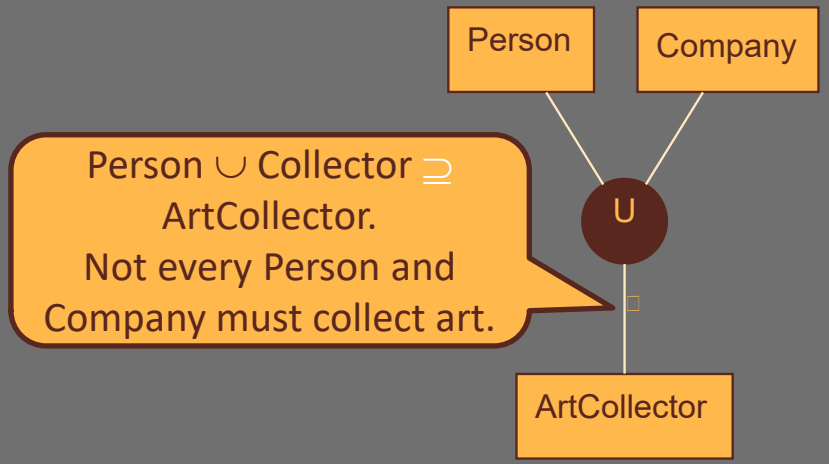
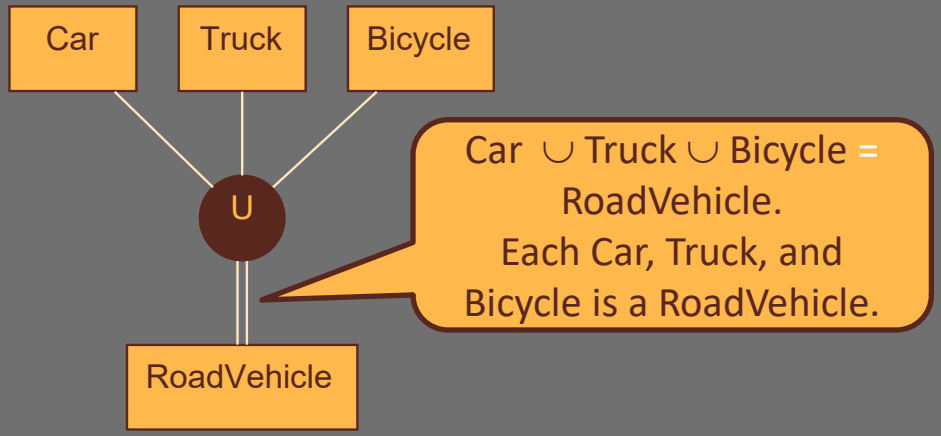
ERD Categories
(aka Union Types)

Review: multiple inheritance hierarchy



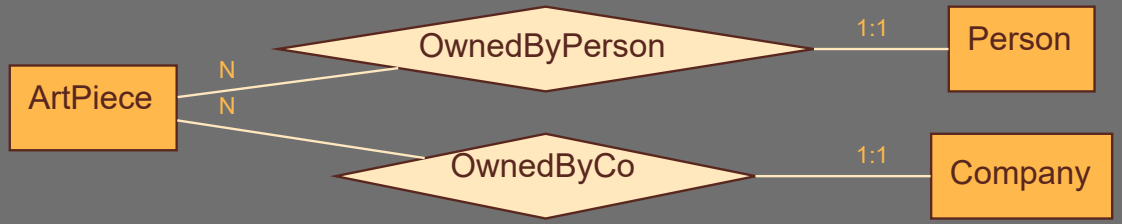
Categories


Person
Company
U
ArtCollector
Person ∪ Collector ⊇ ArtCollector.
Not every Person and Company must collect art.


Car
Truck
Bicycle
U
RoadVehicle
Car ∪ Truck ∪ Bicycle = RoadVehicle.
Each Car, Truck, and Bicycle is a RoadVehicle.

The need for categories

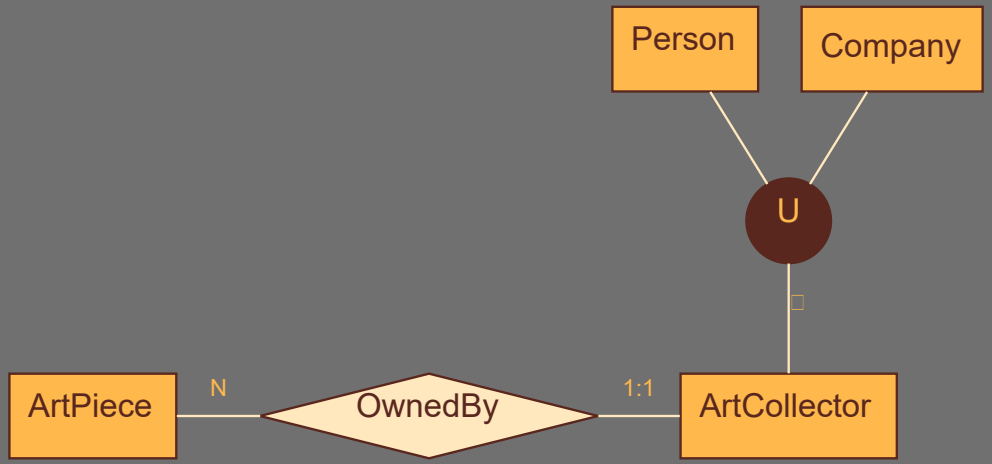
Goal: “Every art piece is owned by a person or company.”



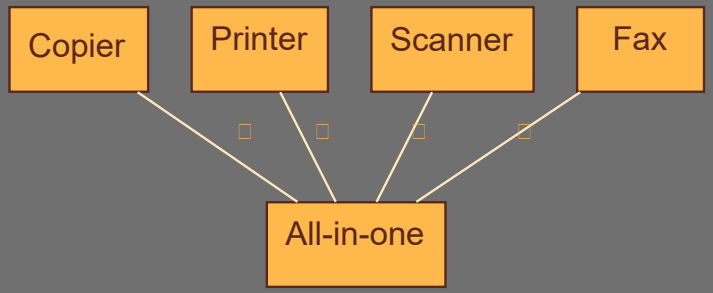
What is the problem?

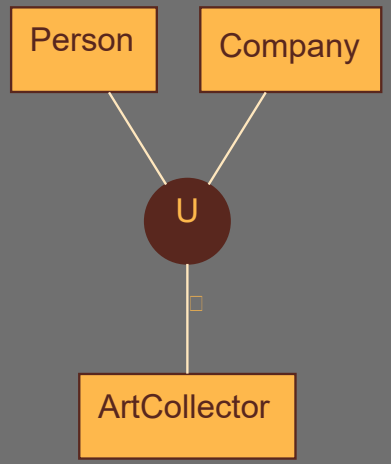
Solution with categories

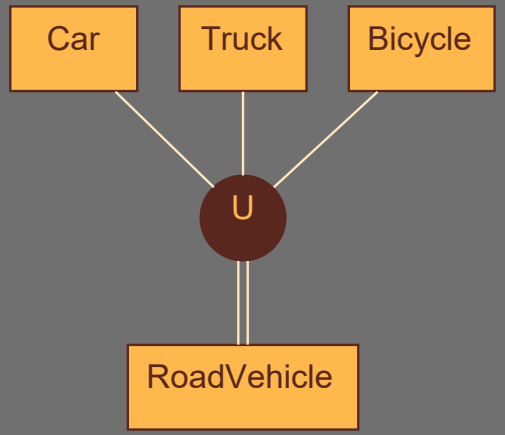
Goal: “Every art piece is owned by a person or company.”



Comparing options for multiple superclasses







Each subclass entity belongs to & inherits from all superclasses.
Same PKs.

Each subclass entity belongs to & inherits from the appropriate one superclass.
Different PKs.

Categories / Union Types

Union of superclasses instead of subclasses.

Subclass entity is in one of the superclasses.

Inherits attributes from that superclass.